

AI-Driven OSINT Threat Monitoring and Risk Assessment: A Guardian Open Platform Case Study

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Abstract

Open-Source Intelligence (OSINT) collection and analysis are central to proactive cyber threat monitoring, yet the manual curation of news and advisory feeds cannot scale. This report presents a prototype that uses Artificial Intelligence (AI) to support an OSINT pipeline. The pipeline ingests publicly available news articles from the Guardian Open Platform Application Programming Interface (API), applies multi-stage Natural Language Processing (NLP) to classify threats and extract key entities, and computes a composite risk score to prioritise intelligence items. The system combines a keyword-based threat taxonomy with a zero-shot Natural Language Inference (NLI) classifier and Valence Aware Dictionary and sEntiment Reasoner (VADER) sentiment analysis, producing ranked threat reports without requiring labelled training data. Across a six-month corpus of 1,228 unique articles, the pipeline produces interpretable per-category intelligence briefs, shows approximately 58% agreement between keyword and NLI classification on the top 50 articles, and reports a mean risk score of 5.09 out of 10, with 63 Critical and 290 High severity items. The results also highlight the main constraints of the prototype: single-source coverage, English-only processing, and hand-tuned scoring weights.

Index Terms

Open-Source Intelligence, threat intelligence, Natural Language Processing, Named Entity Recognition, zero-shot classification, risk scoring, cyber security, Guardian Open Platform.

I. INTRODUCTION

THE volume of publicly available threat-relevant information has grown, spanning news outlets, government advisories, social media, and technical vulnerability databases [1], [2]. Analysts monitoring these channels face a severe bottleneck: filtering signal from noise in near-real time is cognitively demanding and error-prone [3]. Artificial Intelligence (AI) methods, particularly Natural Language Processing (NLP), can support collection, triage, and risk scoring while preserving the interpretability required by operational security teams.

This report presents a prototype that combines standard NLP and classification methods within an Open-Source Intelligence (OSINT) pipeline. The system uses a single open news source, applies several processing stages (cleaning, Named Entity Recognition (NER), classification, and scoring), and ranks the highest-risk articles for analyst review. The emphasis is on design choices and technical trade-offs rather than production hardening. To support reproducibility, the accompanying public GitHub repository includes the notebook, source code, and the configuration to regenerate all artefacts: <https://github.com/sinaebrahimi/threat-monitoring-and-risk-mgmt-prototype>.

A. Contributions

- 1) A complete OSINT ingestion and enrichment pipeline that uses freely available Application Programming Interface (API) access and CPU-only models, demonstrated on 1,228 deduplicated articles spanning six months.
- 2) A two-stage threat classifier that combines fast keyword rules with zero-shot Natural Language Inference (NLI) and requires no labelled training data.
- 3) A composite and interpretable risk-scoring formula that integrates keyword density, sentiment, recency, and entity richness across five threat categories.
- 4) A critical discussion of limitations, ethical considerations, and deployment steps for multi-source and analyst-in-the-loop extensions.

II. RELATED WORK

Early OSINT systems relied on keyword matching and Really Simple Syndication (RSS) aggregation [4]. Recent NER methods based on Bidirectional Encoder Representations from Transformers (BERT) [5] have improved entity extraction from unstructured text. Platforms such as *Open Cyber Threat Intelligence (OpenCTI)* [6] and *Malware Information Sharing Platform (MISP)* [7] formalise threat intelligence sharing, but still rely on analysts to populate structured records, preserving a manual bottleneck.

Zero-shot text classification [8] uses NLI models to score label hypotheses without task-specific training, making it suitable for rapid triage when labelled security data are scarce. Risk quantification approaches range from Common Vulnerability Scoring System (CVSS)-style ordinal scales [9] to learned regression models trained on labelled incident datasets. This report draws on these strands while remaining lightweight: the objective is a prototype that practitioners can inspect and tune, rather than a black-box model optimised only for benchmark accuracy.

III. SYSTEM ARCHITECTURE

Fig. 1 shows the five-stage pipeline. Each stage is implemented as an independent Python module, so components can be swapped or upgraded without rewriting the rest of the system. The pipeline uses three main data formats: JSON, HTML, and CSV.

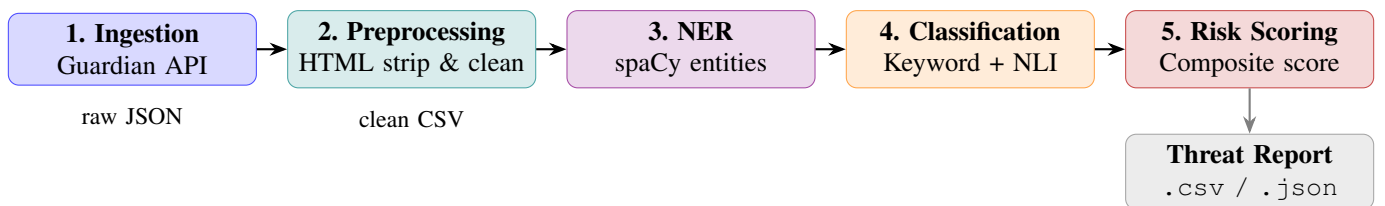


Fig. 1: Five-stage pipeline from API ingestion to ranked threat report.

Stage 1 (Ingestion): A `GuardianCollector` class queries the Guardian `/search` endpoint across 15 threat-relevant terms over a six-month date window. Articles are deduplicated by Guardian content ID and persisted as raw JSON, supporting offline re-runs without repeated API calls.

Stage 2 (Preprocessing): Article bodies are returned as HTML; `BeautifulSoup`, with an `lxml` backend, strips the markup to plain text. A `search text` field is then constructed by concatenating the title

(repeated three times for downstream emphasis), the standfirst, and the body, providing a single string for all NLP stages.

Stage 3 (Named Entity Recognition): spaCy’s `en_core_web_sm` [10] is applied to the first 5000 characters of each article, extracting Organisation (ORG), Geopolitical Entity (GPE), Person (PERSON), Event (EVENT), and Nationalities or Religious or Political Groups (NORP) spans.

Stage 4 (Classification): A two-stage classifier (Sec. V-B) assigns each article to one of five threat categories: *Cyber*, *Geopolitical*, *Terrorism*, *Disinformation*, and *Critical Infrastructure*.

Stage 5 (Risk Scoring): Four normalised component scores are combined into a composite risk score (Sec. VI), scaled to $[0, 10]$ and mapped to four severity labels.

IV. DATA COLLECTION

A. Guardian Open Platform

The Guardian Open Platform provides free developer-tier access to article full text, metadata, and taxonomic tags via a Representational State Transfer (REST) API [11]. The `/search` endpoint accepts keyword queries, date ranges, section filters, and a `show-fields` parameter that controls the returned fields. Full article body text is available at no cost, making this a high-quality and reproducible data source for OSINT prototyping.

B. Corpus Construction

Table I lists the 15 threat-oriented query terms. For each term, up to three pages of 50 results (150 articles per term) are retrieved, covering a 180-day period from December 2025 to June 2026. After deduplication using Guardian content IDs, the corpus contains 1,228 unique articles. A 0.25-second inter-request delay ensures compliance with the developer-tier rate limit. Articles with fewer than 50 body characters, typically gallery or video-primary content, are removed during preprocessing; in this corpus only a single article was affected.

TABLE I: Threat query terms used to build the corpus

Category	Query terms
Cyber	cyberattack, ransomware, data breach, malware, zero-day vulnerability, phishing, supply chain attack , Distributed Denial of Service (DDoS) attack
Geopolitical	espionage, nation state hacking, cyber warfare
Terrorism	terrorism
Disinformation	disinformation, information warfare
Infrastructure	critical infrastructure attack

C. Dataset Characteristics

The resulting category distribution is shown in Fig. 2. Terrorism is the largest category, followed by Cyber, Critical Infrastructure, Geopolitical, Unknown, and Disinformation. This pattern reflects both the relative volume of Guardian reporting on each topic and the number of query terms per category. The most represented sections are `world`, `us-news`, and `australia-news`, with a mean article word count of approximately 1,700.

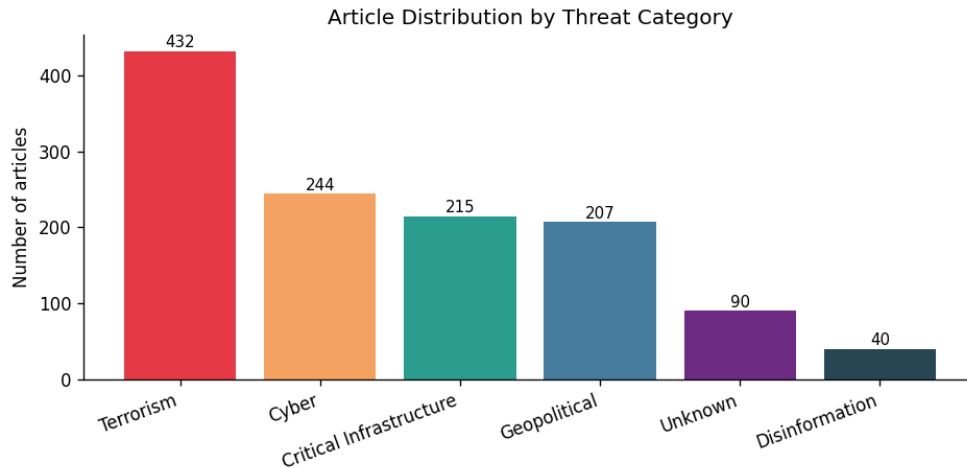


Fig. 2: Distribution of 1,228 articles across the five threat categories plus Unknown, based on keyword classification.

V. METHODOLOGY

A. Named Entity Recognition

spaCy’s statistical NER pipeline identifies named spans within article text and assigns them to entity types. The `en_core_web_sm` model (12 MB, CPU-only) achieves an F1 score of approximately 0.85 on OntoNotes [10], [12]. To bound runtime, processing is limited to the first 5000 characters of each article; the title and standfirst, typically the most informationally dense fields, always fall within this window. Entity extraction serves two purposes: (i) it provides structured metadata for the threat report by identifying which organisations and countries are implicated in each item; and (ii) it contributes a risk signal, since articles naming more distinct actors tend to describe more concrete, attributable threats.

Fig. 3 visualises the most frequent named entities extracted from threat articles. It was generated in the notebook’s visualisation step by concatenating the extracted ORG, GPE, and PERSON fields, replacing pipe delimiters with spaces, and rendering the 120 most frequent terms as a word cloud. Larger terms indicate entities that appear more often in the scored corpus.



Fig. 3: Named entities extracted from threat articles (ORG, GPE, and PERSON).

B. Threat Classification

Classification proceeds in two stages.

Stage 1: Keyword classifier (all 1,228 articles): For each category c , keyword density is

$$K_c = \frac{\sum_{k \in \mathcal{K}_c} \text{count}(k, \hat{t})}{|\hat{t}|} \quad (1)$$

where $\mathcal{K}_c$ denotes category keywords, $\hat{t}$ the lowercased search text, and $|\hat{t}|$ the word count. The label is $c^* = \arg \max_c K_c$, with keyword score $S_{\text{kw}} = \min(K_{c^*}/0.05, 1.0)$ capped at a reference density of 0.05 hits per word.

Stage 2: Zero-shot NLI classifier (top-50 by keyword score): The pipeline uses the NLI cross-encoder model [13], `cross-encoder/nli-MiniLM2-L6-H768`, in zero-shot mode [8]. For each article, the model evaluates each category label as a natural-language hypothesis and returns a confidence distribution. To bound runtime, this stage is restricted to the top-50 articles, requiring approximately two seconds per article on a CPU. The highest-confidence label and score are stored with the keyword result, enabling consistency checks and illustrating how learned models can complement rule-based classification.

C. Sentiment Analysis

Valence Aware Dictionary and sEntiment Reasoner (VADER) [14] produces a compound score $v \in [-1, 1]$ for each article headline and standfirst. The sentiment-risk component is defined as

$$S_{\text{sent}} = \frac{1 - v}{2} \quad (2)$$

so that highly negative text ($v \approx -1$) yields $S_{\text{sent}} \approx 1$, neutral text maps to 0.5, and positive text approaches 0. VADER was chosen over a transformer-based sentiment model because it runs in milliseconds on commodity hardware, requires no additional model download, and produces directly interpretable scores.

VI. RISK ASSESSMENT FRAMEWORK

A. Component Scores

The composite risk score combines four normalised component scores, each bounded in $[0, 1]$:

- 1) **Keyword score** S_{kw} : normalised keyword density (Eq. 1).
- 2) **Sentiment score** S_{sent} : VADER-derived negativity (Eq. 2).
- 3) **Recency score** S_{rec} : exponential decay from the publication date, using a 30-day half-life, $S_{\text{rec}} = \exp\left(-\frac{\ln 2 \cdot \Delta t}{30}\right)$, where Δt is the article age in days.
- 4) **Entity score** S_{ent} : normalised named-entity count, $\min(n_e/n_{\text{max}}, 1.0)$, where n_e is the article's entity count and n_{max} is the maximum entity count in the corpus.

B. Composite Risk Score

The raw, unclipped risk score is

$$\text{Risk} = \left(0.35 \cdot S_{\text{kw}} + 0.25 \cdot S_{\text{sent}} + 0.20 \cdot S_{\text{rec}} + 0.20 \cdot S_{\text{ent}}\right) \times M_c \times 10, \quad (3)$$

where M_c is a category-specific severity multiplier: Terrorism (1.5), Critical Infrastructure (1.4), Cyber (1.3), Geopolitical (1.2), Disinformation (1.0), and Unknown (0.8). These multipliers encode domain knowledge about each category’s typical operational impact; they are deliberate design choices rather than learned parameters. The reported score is then clipped to $[0, 10]$ and mapped to four severity labels (Table II). Because M_c is applied before clipping, the raw score can reach 15.0 when all components are maximal and $M_c = 1.5$, so several items may share the reported ceiling of 10.0.

TABLE II: Severity labels and suggested analyst actions

Label	Score range	Suggested action
Low	[0, 3)	Log; review weekly
Medium	[3, 6)	Monitor; alert on escalation
High	[6, 8)	Prioritise; analyst review within 24 h
Critical	[8, 10]	Immediate escalation

C. Threat Report Output

The pipeline produces two artefacts. The CSV file, `threat_report.csv` (555 KB), lists all articles in descending risk-score order, including category, severity label, key entities, and URL. The JSON file, `threat_report.json` (832 KB), stores the same records for downstream dashboards or Security Information and Event Management (SIEM) platforms. A structured set of per-category *intelligence briefs* (step 7c in the `threat_monitoring.ipynb` notebook) extracts the two highest-scoring articles per threat type and summarises title, score, date, key organisations, and locations.

VII. PROTOTYPE RESULTS

This section summarises the prototype’s empirical outputs: corpus composition, risk-score distributions, classifier agreement, and sentiment trends.

A. Corpus Overview

The corpus composition in Fig. 2 provides context for the downstream scoring results. In particular, the prevalence of articles classified as Terrorism ($\sim 35\%$) interacts with the category multiplier ($M_c = 1.5$), amplifying scores for high-keyword-density matches in that category.

B. Risk Score Distribution

The composite score distribution is right-skewed: 60.8% of articles fall in the Medium range ($[3, 6)$), with 23.6% in High, 5.1% in Critical, and 10.4% in Low. The corpus mean is 5.09 out of 10. Fig. 4 shows the histogram. The highest-scoring articles (scores of 10.0) cover the Golders Green terrorist attack and its immediate aftermath in the UK [15]. These articles combine high keyword density, strong sentiment negativity, and a short elapsed time since publication (S_{rec} in (3)). This ranking follows directly from the scoring formula. Table III lists the five highest-ranked items; all are recent terrorism stories whose raw scores exceed the cap and tie at 10.0, reflecting the combined effect of high keyword density, strong negative sentiment, recency, and the *Terrorism* multiplier ($M_c = 1.5$).

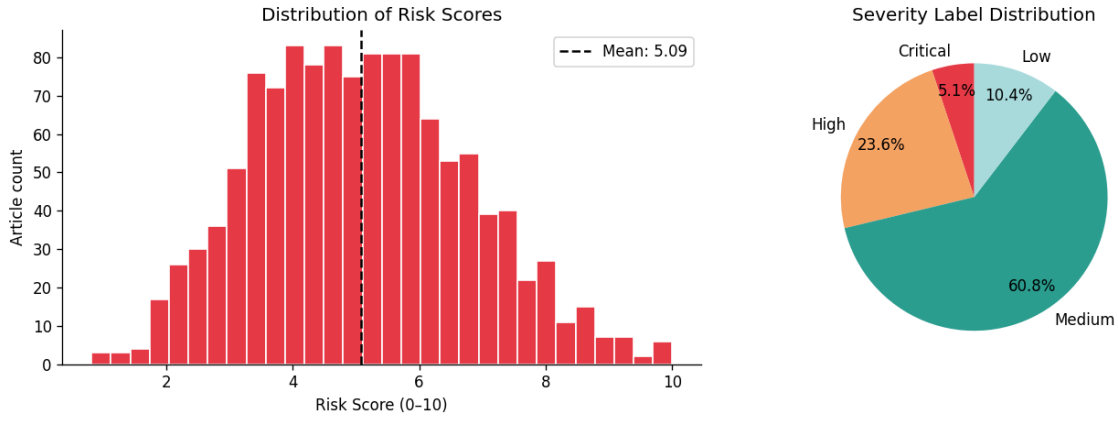


Fig. 4: Risk score distribution across the corpus.

TABLE III: Top five threats by composite risk score.

#	Headline (truncated)	Category	Score
1	UK terrorism threat level raised to “severe” after Golders Green attack [15]	Terrorism	10.0
2	Terror survivors’ group writes letter in support of UK Jews	Terrorism	10.0
3	Canberra teenager charged over allegedly planning ACT terror attack	Terrorism	10.0
4	Stafford man pleads guilty to terrorism offences over ‘killing migrants’ posts	Terrorism	10.0
5	US charges man with plotting Iran-directed attacks on Jews in London and New York	Terrorism	10.0

C. Risk by Category

Fig. 5 compares composite risk scores across the five threat categories plus Unknown. The x-axis shows the assigned category, and the y-axis shows the final clipped risk score. The boxplot shows that *Terrorism* has the highest median score and widest upper tail, followed by *Critical Infrastructure*; *Disinformation* and *Unknown* are concentrated at lower scores. This pattern reflects both article-level signals and the category multipliers in Eq. 3, so it should be interpreted as a prioritisation output rather than as ground-truth severity.

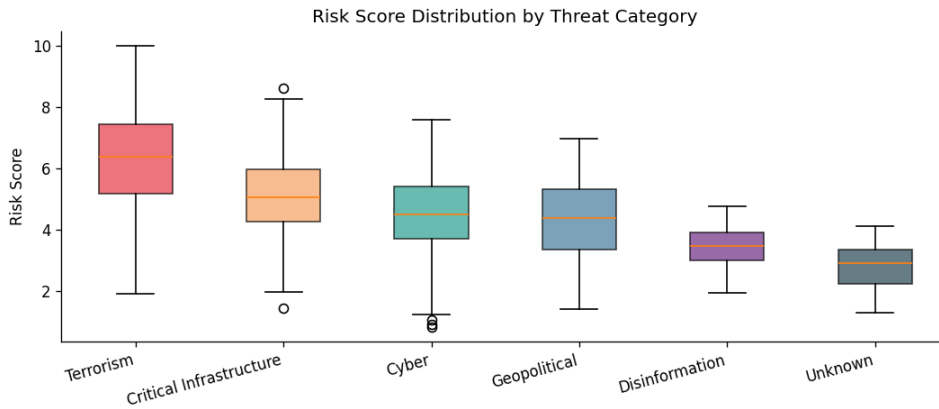


Fig. 5: Composite risk score by threat category.

D. Classifier Comparison

For the top 50 articles by keyword score, the keyword and NLI classifiers agree approximately 58% of the time. Disagreements arise most often when the keyword classifier assigns *Terrorism*, driven by its broad *attack/terror* vocabulary, while the zero-shot model assigns *Geopolitical*, typically with low confidence. These cases show that the two stages are complementary: the keyword classifier provides fast, fully deterministic triage, whereas the NLI stage captures semantic context that keyword density alone can miss. A known limitation is that articles covering the war in Ukraine are sometimes scored as high-risk Terrorism items because they contain dense attack-related vocabulary; more granular subcategories would reduce this artefact in a production system.

E. Risk versus Sentiment

Fig. 6 shows the relationship between composite risk score and VADER sentiment negativity across the corpus, with points coloured by threat category. High-risk articles cluster at higher negativity values, but the spread is substantial, confirming that sentiment is a meaningful but non-dominant signal. Variation in recency and keyword density introduces this spread, which is the intended behaviour of the composite formula.

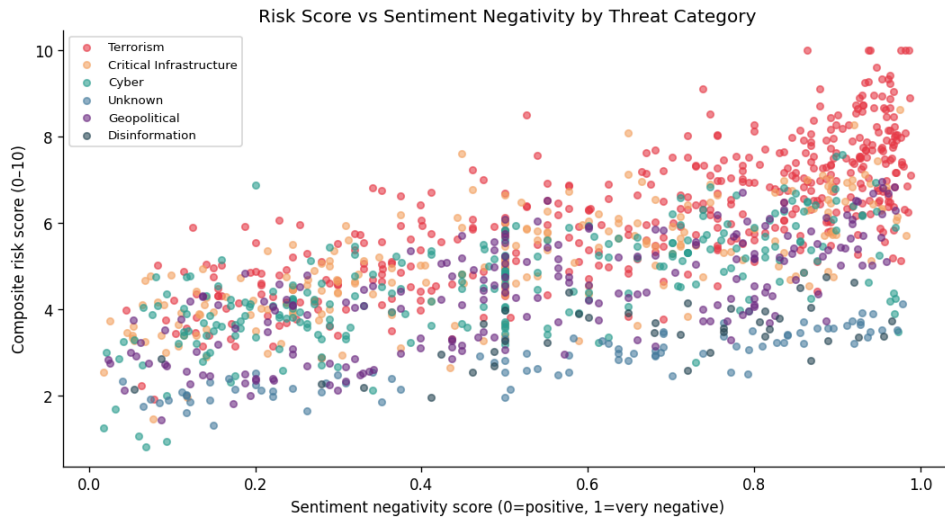


Fig. 6: Risk score versus VADER sentiment negativity across the corpus.

VIII. KEY CONSIDERATIONS

A. Data Quality and Limitations

The prototype relies on a single English-language news outlet. The Guardian’s editorial decisions determine what is published; emerging threats first reported in non-English sources or specialist technical forums, such as threat intelligence blogs, vulnerability disclosure lists, and dark-web forums, are not captured by the system. A production pipeline should integrate multiple independent sources, including Cybersecurity and Infrastructure Security Agency (CISA) advisories and Common Vulnerabilities and Exposures (CVE) feeds [9], and apply cross-lingual NLP.

Keyword classifiers are inherently vocabulary-bound: novel terminology, such as a newly coined attack name or a zero-day before public disclosure, will not match until the taxonomy is updated. Conversely, broad terms such as `attack` can generate false positives, drawing sport, entertainment, or accident reporting into high-severity categories; for example, references to the band *Massive Attack* can be mis-scored as *Terrorism*. These errors reinforce the need for context-aware filtering and human review of high-severity items. Zero-shot NLI can partially mitigate such cases but remains constrained by its training distribution. The scoring weights are hand-tuned rather than data-driven; without labelled ground truth, there is no objective basis for claiming that they are optimal, and different weighting configurations would yield different rankings.

B. Ethical and Legal Considerations

All data is retrieved under The Guardian’s developer API terms, which permit non-commercial use. The prototype processes no personal data beyond article bylines. However, automated threat labelling at scale can generate false positives that, if acted upon without human review, could harm individuals or organisations. A human-in-the-loop review step is therefore essential before any score-driven operational action is taken. The system should not be used to monitor individuals, and any production deployment should include bias audits to ensure that no group is systematically over-represented in high-severity labels.

C. Scalability and Deployment

The prototype runs on a standard laptop without GPU acceleration. Production deployment would require three architectural changes: (i) replacing manual notebook execution with a scheduled ingestion pipeline, such as Apache Airflow; (ii) replacing CSV files with a vector database, such as Weaviate or Pinecone, to support semantic search and similarity-based deduplication as the corpus grows; and (iii) replacing the small, general-purpose models with fine-tuned cybersecurity NER and classification models. Table IV summarises these transitions.

TABLE IV: Scalability steps: prototype versus production

Component	Prototype	Production
Ingestion	Manual notebook run	Airflow Directed Acyclic Graph (DAG), multi-source
Storage	JSON + CSV files	Vector database (Weaviate / Pinecone)
NER model	spaCy sm (12 MB)	Fine-tuned cybersecurity NER model
Classifier	Keyword + zero-shot NLI sample	Fine-tuned transformer
Risk scoring	Hand-tuned weights	Active-learning feedback
Output	Static CSV/JSON	Live Streamlit dashboard

A later version could adopt an *agentic* architecture with per-source collector agents for monitoring news, advisories, CVE feeds, and other OSINT channels; a triage/scoring agent for deduplication, categorisation, and risk ranking; and an analyst-feedback agent for capturing review decisions and updating scoring parameters. This would broaden source coverage while keeping analyst oversight central.

IX. CONCLUSION AND FUTURE WORK

This report has presented a prototype OSINT threat monitoring system that runs an end-to-end pipeline from API ingestion to ranked threat reports, using only CPU-friendly, freely available NLP tools. Across 1,228 articles, the pipeline achieves a corpus mean risk score of 5.09, identifies Critical items with scores up to 10.0, and produces structured intelligence briefs across five threat categories. The two-stage classifier requires no labelled data, and the composite risk score is fully interpretable. The main limitations are single-source coverage, English-only processing, and hand-tuned weights.

Future work should focus on: (i) integrating CISA advisories and CVE feeds for enriched technical context; (ii) mapping extracted entities to the MITRE Adversarial Tactics, Techniques, and Common Knowledge (ATT&CK) framework [16] for structured threat attribution; (iii) building a knowledge graph of threat actors, tools, and campaigns; (iv) applying active learning with analyst feedback to refine classification and scoring weights; and (v) developing a real-time Streamlit dashboard as the operational interface. Over time, this could evolve into a fully agentic architecture capable of autonomous multi-source OSINT collection and triage.

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